

Understanding Multilateral Tariffs and Exchange Rates

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1 Introduction

The conventional wisdom from a standard two-country model of international trade is that a tariff shifts demand toward domestic goods, improving the tariffing country's trade balance and appreciating its currency under the Marshall–Lerner condition. With more than two countries, however, a tariff on one partner also reallocates expenditure across foreign suppliers, so bilateral exchange rates need not move in the same direction. When differentiated tariffs are imposed on multiple foreign suppliers, these effects interact and become harder to characterize. This paper revisits the conventional two-country result and develops a framework for understanding how tariffs affect bilateral and effective exchange rates in a multilateral setting.

In a general N -country trade environment, we show that the log-linearized exchange-rate response to tariffs can be represented as $(-J)^{-1}F$, where F gives the trade-balance effects of tariffs at unchanged exchange rates and J is the Jacobian of trade balances with respect to exchange rates. This representation separates the initial trade-balance effects of the tariffs from the exchange-rate adjustment required to restore equilibrium. We introduce a multilateral Marshall–Lerner (MML) condition on J to characterize this adjustment and show that, under partner symmetry, the response to any tariff vector decomposes into average protection and relative treatment. Average protection determines the change in the tariffing country's nominal effective exchange rate, while relative treatment determines the distribution of bilateral exchange rates around that effective rate.

To quantify these effects in a parsimonious framework, we develop a multilateral trade model with nested-CES preferences. The exchange-rate responses are governed by three margins built from the underlying elasticities: a domestic diversion margin ρ_D , which combines substitution toward the domestic tradable, governed by the domestic-import elasticity η , with the income effect of rebated tariff revenue on non-tradables; substitution across foreign suppliers, governed by the elasticity ρ ; and a multilateral exchange-rate adjustment margin $\mathcal{M}_N$. For a unilateral tariff, we show that the tariffing country always appreciates against the targeted country, while its exchange rate against an untariffed bystander can depreciate when substitution across foreign suppliers is sufficiently strong relative to the domestic diversion margin. We then consider bilateral retaliation and retaliation by all partners and show that retaliation lowers the threshold for depreciation against bystander countries by adding a relative-treatment effect that offsets the average-protection force toward appreciation. Finally, we relate these results to the preferential-trade-agreement literature on third-country terms-of-trade effects. We show that standard competing-exporter models can be

understood as deliberately abstracting from some of the domestic-substitution and income channels present in our model in order to isolate substitution across foreign suppliers.

We abstract from several features that also matter for the exchange-rate effects of tariffs. Some are primarily relevant for short-run adjustment, including nominal and real adjustment frictions, trade imbalances, and capital flows. Others are likely to remain important in the long run, including productivity growth, capital and intermediate goods, supply chains, and re-exporting. We also abstract from risk, monetary and fiscal policy, and financial markets.

2 Tariffs and Exchange-Rate Adjustment

This section develops the general N -country exchange-rate adjustment representation, introduces its multilateral Marshall–Lerner condition, and derives the implications of partner symmetry for a general tariff-change vector. We begin with the bilateral case and then extend the same logic to a general differentiable trade system.

2.1 The bilateral case

Consider first two countries, A and B . Let e_{AB} denote units of A 's currency per unit of B 's, so that a rise in e_{AB} is a depreciation of A . Country A levies a tariff τ on imports from B . Both the tariff and the exchange rate affect relative consumer prices and hence A 's trade balance. The balanced-trade equilibrium is therefore defined by

$$\text{TB}_A(e_{AB}, \tau) = 0.$$

Write $\nu \equiv \log e_{AB}$. Linearizing the trade-balance condition around an equilibrium gives

$$d\text{TB}_A = \text{TB}_{A\nu} d\nu + \text{TB}_{A\tau} d\tau, \quad \text{TB}_{A\nu} \equiv \frac{\partial \text{TB}_A}{\partial \nu}, \quad \text{TB}_{A\tau} \equiv \frac{\partial \text{TB}_A}{\partial \tau} \Big|_{\nu \text{ fixed}}. \quad (1)$$

Suppose that $\text{TB}_{A\tau} > 0$, i.e., the tariff improves A 's trade balance at unchanged exchange rates. This is the standard case in which the tariff raises the domestic price of imports and shifts expenditure away from foreign goods, reducing import demand relative to exports. Restoring balance then requires

$$d\nu^* = -\frac{\text{TB}_{A\tau}}{\text{TB}_{A\nu}} d\tau. \quad (2)$$

For a tariff increase $d\tau > 0$, this is an appreciation if and only if $\text{TB}_{A\nu} > 0$, that is, if and only if a depreciation improves A 's trade balance. At a balanced-trade baseline,

$$\text{TB}_{A\nu} = X(\varepsilon_X + \varepsilon_M - 1),$$

where X is the value of exports, and ε_X and ε_M are the export and import demand elasticities, respectively. Hence a depreciation improves A 's trade balance under the bilateral Marshall–Lerner condition

$$\varepsilon_X + \varepsilon_M > 1. \quad (3)$$

Figure 1 illustrates the logic of Equations (1)–(2). The tariff first changes the trade balance at unchanged exchange rates, and under the bilateral Marshall–Lerner condition (3), the exchange rate then appreciates to restore balance.

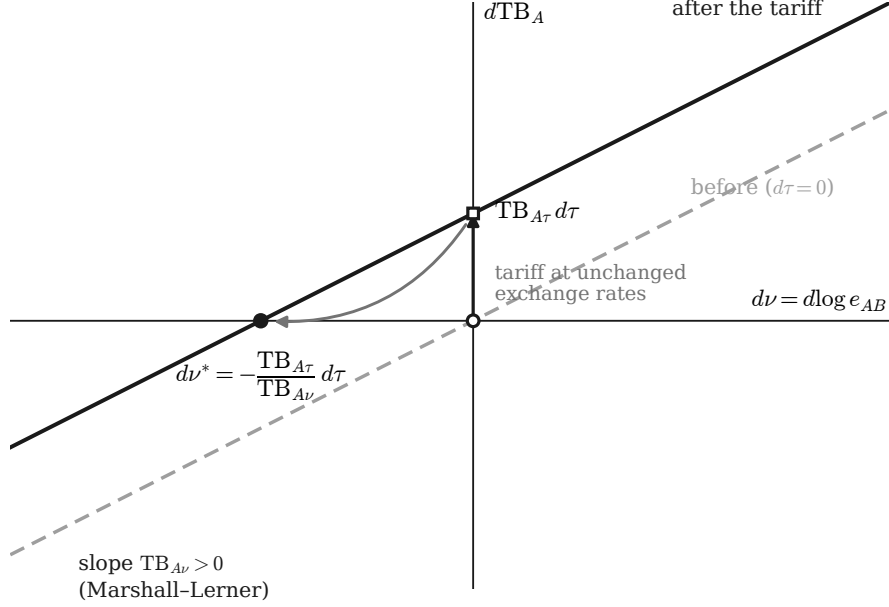


Figure 1: Bilateral tariff and exchange-rate adjustment under $TB_{A\tau} > 0$ and $TB_{A\nu} > 0$.

2.2 The local disturbance–adjustment representation

Now consider a differentiable N -country trade environment. Taking A 's currency as the numeraire, the $N - 1$ bilateral exchange rates e_{Aj} , $j \neq A$, span all exchange-rate changes, and Walras' law leaves $N - 1$ independent trade-balance conditions. We therefore characterize the balanced-trade equilibrium by

$$TB_i(\{e_{Aj}\}_{j \neq A}, \{\tau_{jk}\}_{j \neq k}) = 0, \quad i \neq A.$$

Stack the log exchange rates as $\boldsymbol{\nu} \equiv (\nu_j)_{j \neq A}$, with $\nu_j \equiv \log e_{Aj}$, and the bilateral tariff changes as $d\boldsymbol{\tau} \equiv (d\tau_{jk})_{j \neq k}$. Linearizing the $N - 1$ trade-balance conditions gives

$$J d\boldsymbol{\nu} + G d\boldsymbol{\tau} = 0, \quad J_{il} \equiv \frac{\partial TB_i}{\partial \nu_l}, \quad G_{i,(jk)} \equiv \frac{\partial TB_i}{\partial \tau_{jk}} \Big|_{\boldsymbol{\nu} \text{ fixed}}. \quad (4)$$

Now separate a general tariff change into its direction and scale. Let $\mathbf{a}$ denote the direction of the tariff change, normalized so that $\|\mathbf{a}\|_2 = 1$, and let $ds \geq 0$ denote its scale, so that

$$d\boldsymbol{\tau} = \mathbf{a} ds.$$

The vector $\mathbf{a}$ therefore describes which bilateral tariffs change and in what proportions, while

$ds = \|d\tau\|_2$ gives the magnitude of the tariff change. For a given direction $\mathbf{a}$, define

$$\mathbf{F} \equiv G \mathbf{a}, \quad (5)$$

where $\mathbf{F}$ is the vector of trade-balance disturbances generated by a unit increase in the scale of the tariff change at unchanged exchange rates. Substituting $d\tau = \mathbf{a} ds$ into (4) gives

$$J d\boldsymbol{\nu} + \mathbf{F} ds = 0. \quad (6)$$

Hence the marginal exchange-rate response along tariff-change direction $\mathbf{a}$ is

$$\frac{d\boldsymbol{\nu}}{ds} = (-J)^{-1} \mathbf{F}. \quad (7)$$

Equation (7) is the multilateral analogue of the bilateral adjustment in Equation (2) (setting $N = 2$, $\mathbf{a} = 1$, and $ds = d\tau$). This expression shows that the exchange-rate response to tariffs can be separated into two parts. The vector $\mathbf{F}$, and more generally G , describes the initial trade-balance effects of tariff changes at unchanged exchange rates, including substitution and income effects. The matrix J describes how trade balances respond to exchange-rate changes, while $(-J)^{-1}$ maps the initial disturbance into the exchange-rate adjustment required to restore equilibrium.

2.3 The multilateral Marshall–Lerner condition

In the bilateral case, the requirement that exchange-rate adjustment stabilizes the trade balance is the scalar condition $-J = \text{TB}_{A\nu} > 0$, which has a straightforward interpretation. In the multilateral case, $-J$ relates a vector of exchange-rate changes to the trade-balance disturbances that those changes can offset, while $(-J)^{-1}$ gives the exchange-rate changes required to offset a given vector of disturbances. A natural analogue of the bilateral Marshall–Lerner condition is therefore that this adjustment be well defined and have the stabilizing sign.

Definition 2.1 (Multilateral Marshall–Lerner condition). *The adjustment mechanism satisfies the multilateral Marshall–Lerner condition (MML) if*

$$J_{il} \geq 0 \quad (l \neq i), \quad (-J)^{-1} \text{ exists and satisfies } (-J)^{-1} \geq 0. \quad (8)$$

Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, a positive trade-balance disturbance in any country makes a weakly positive contribution to every bilateral exchange-rate response. By (7), writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_j \vartheta_{ij} F_j, \quad (9)$$

where ϑ_{ij} measures the response of the A – i bilateral exchange rate required to offset a unit trade-balance disturbance in country j , with a positive response denoting a depreciation of A against i . A uniformly signed $\mathbf{F}$ therefore cannot generate exchange-rate responses of the opposite sign.

When $\mathbf{F}$ has mixed signs, the response of any individual exchange rate depends on the full vector of trade-balance effects rather than on the corresponding element F_i alone.

The MML condition stated directly in terms of $(-J)^{-1}$ is somewhat abstract. A more intuitive sufficient condition can be stated in terms of gross substitutability. To express this condition without choosing a numeraire, extend the notation and let ν_i , $i = 1, \dots, N$, denote the log value of currency i in a common unit of account, so that

$$d \log e_{ij} = d\nu_j - d\nu_i,$$

and let $\tilde{J}_{il} \equiv \partial \text{TB}_i / \partial \nu_l$ be the full $N \times N$ Jacobian. Homogeneity of each trade balance in currency values implies, by Euler's theorem, that

$$\sum_l \tilde{J}_{il} = \text{TB}_i,$$

so at a balanced equilibrium

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i.$$

Separately, $\sum_i \text{TB}_i \equiv 0$ by Walras' law implies

$$\sum_i \tilde{J}_{il} = 0 \quad \forall l.$$

Thus,

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i, \quad \sum_i \tilde{J}_{il} = 0 \quad \forall l. \quad (10)$$

The row identity pins down each diagonal entry of $\tilde{J}$ from the off-diagonal entries, so the relevant sign restriction can be stated entirely in terms of these cross-effects.

Assumption 1 (Gross substitutability of trade balances). $\tilde{J}_{il} \geq 0$ for all $l \neq i$.

An appreciation of currency l raises the price of l 's goods relative to those produced elsewhere. Assumption 1 requires that this weakly improve every other country's trade balance once the associated income and expenditure effects are included. The intuition is the same as in the bilateral Marshall–Lerner condition: exchange-rate movements operate in the direction needed to offset trade-balance disturbances. Together with the row identity in (10), this sign structure makes the reduced matrix weakly diagonally dominant; the additional connectivity conditions in Lemma 2.2 ensure that $-J$ is a nonsingular M-matrix.

Lemma 2.2 (Sufficiency of gross substitutability). *Suppose each trade balance is homogeneous of degree one in currency values. Under Assumption 1, with J irreducible and $\tilde{J}_{iA} > 0$ for some i , MML (8) holds.*

Thus, under gross substitutability and the conditions of Lemma 2.2, the multilateral Marshall–Lerner condition follows from economically interpretable cross-effects of exchange rates on trade balances. The proof is given in Appendix F.

2.4 Symmetric partners

Now restrict attention to tariff changes imposed by A , and write $\mathbf{t} = (t_j)_{j \neq A}$, with $t_j \equiv d\tau_{Aj}$. Restricting G to these $N - 1$ tariff changes, let G henceforth denote the resulting $(N - 1) \times (N - 1)$ matrix. Suppose that A 's trading partners are symmetric, so that relabeling them leaves the reduced system unchanged:

$$PJP' = J, \quad PGP' = G$$

for every permutation matrix P of the $N - 1$ partners.

Under partner symmetry, all partners are interchangeable except for how they are treated by A 's tariff changes. It is therefore natural to separate the part of the tariff change that is common across partners from the part that distinguishes among them. We refer to these as *average protection* and *relative treatment*, respectively. Decompose

$$\mathbf{t} = \bar{t} \mathbf{1} + \tilde{\mathbf{t}}, \quad \bar{t} \equiv \frac{1}{N-1} \mathbf{1}' \mathbf{t}, \quad \mathbf{1}' \tilde{\mathbf{t}} = 0. \quad (11)$$

Componentwise,

$$t_j = \bar{t} + (t_j - \bar{t}), \quad \tilde{t}_j \equiv t_j - \bar{t}. \quad (12)$$

Thus $\bar{t} \mathbf{1}$ is the common component of the tariff change across partners, while $\tilde{\mathbf{t}}$ records each partner's treatment relative to that common component.

Partner symmetry also gives these two components a simple eigenvalue interpretation. For G , the input is a vector of tariff changes and the output is the resulting vector of trade-balance disturbances at unchanged exchange rates. A common tariff change is therefore an eigenvector of G , and every zero-sum relative-treatment change is an eigenvector with a common eigenvalue:

$$G\mathbf{1} = g_S \mathbf{1}, \quad Gv = g_D v \quad \text{for any } v \text{ such that } \mathbf{1}' v = 0.$$

Thus g_S measures the trade-balance disturbance generated by average protection, while g_D measures the disturbance generated by relative treatment. Similarly, for $-J$, the input is a vector of exchange-rate changes and $-J d\nu$ is the trade-balance disturbance that those changes can offset. Symmetry implies

$$(-J)\mathbf{1} = j_S \mathbf{1}, \quad (-J)v = j_D v \quad \text{for any } v \text{ such that } \mathbf{1}' v = 0,$$

so j_S and j_D measure the strength of exchange-rate adjustment for common and relative movements, respectively.

Now define

$$d\nu_A^E \equiv \frac{1}{N-1} \sum_{j \neq A} d\nu_j,$$

which denotes the equally weighted average of A 's bilateral exchange-rate responses. At a symmetric baseline with equal bilateral trade weights, this coincides with the change in A 's nominal effective exchange rate in (21). The following result combines the tariff disturbances and exchange-rate adjustment along the two eigen-directions.

Proposition 2.3 (Average protection and relative treatment). *Suppose A 's partners are symmetric and $N \geq 3$. Let g_S, g_D denote the eigenvalues of G for common and zero-sum tariff changes, and let j_S, j_D denote the corresponding eigenvalues of $-J$ for common and zero-sum exchange-rate changes. If $j_S, j_D \neq 0$, the response to any tariff-change vector $\mathbf{t}$ is*

$$d\boldsymbol{\nu} = \lambda_A \bar{t} \mathbf{1} + \lambda_R \tilde{\mathbf{t}}, \quad \lambda_A \equiv \frac{g_S}{j_S}, \quad \lambda_R \equiv \frac{g_D}{j_D}. \quad (13)$$

Hence

$$d\nu_A^E = \lambda_A \bar{t}, \quad d\nu_j - d\nu_A^E = \lambda_R \tilde{t}_j, \quad d \log e_{jk} = \lambda_R (t_k - t_j), \quad (14)$$

so average protection determines the common exchange-rate response, while relative treatment determines bilateral deviations and cross rates. Furthermore, MML (8) is equivalent to

$$j_S > 0, \quad j_D \geq j_S. \quad (15)$$

The eigenvalue structure follows directly from partner symmetry. Any matrix invariant to relabeling the partners has the form $aI + b\mathbf{1}\mathbf{1}'$. Since $\mathbf{1}'\mathbf{1} = N - 1$ and $\mathbf{1}'v = 0$ for any zero-sum vector v ,

$$(aI + b\mathbf{1}\mathbf{1}')\mathbf{1} = (a + (N - 1)b)\mathbf{1}, \quad (aI + b\mathbf{1}\mathbf{1}')v = av.$$

Thus the common direction has eigenvalue $a + (N - 1)b$, while every zero-sum direction has eigenvalue a . For $-J$ and G ,

$$\begin{aligned} -J = a_J I + b_J \mathbf{1}\mathbf{1}' : \quad j_S &= a_J + (N - 1)b_J, \quad j_D = a_J, \\ G = a_G I + b_G \mathbf{1}\mathbf{1}' : \quad g_S &= a_G + (N - 1)b_G, \quad g_D = a_G. \end{aligned} \quad (16)$$

Applying these eigenvalues to $\mathbf{t} = \bar{t} \mathbf{1} + \tilde{\mathbf{t}}$ gives (13). Because relative treatment sums to zero, it has no effect on the equally weighted average of the bilateral exchange-rate responses, giving the first equality in (14). The MML characterization in (15) follows from the same structure. Since

$$b_J = \frac{j_S - j_D}{N - 1},$$

$j_D \geq j_S$ is equivalent to $b_J \leq 0$, which gives the required nonnegative off-diagonal elements of J . Together with $j_S > 0$, this also implies $(-J)^{-1} \geq 0$. The formal proof is given in Appendix F.

3 The Nested-CES Model

We now specify an N -country nested-CES model that gives economic structure to the general results of Section 2.

3.1 Production, preferences, and equilibrium

We index countries by $i, j \in \{1, \dots, N\}$. Where appropriate, we also use A for the tariffing country, B for the tariff target, and C for an untariffed bystander. Each country produces a country-specific tradable good and a non-tradable good using labor only, with linear production functions

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i.$$

Normalizing the producer price of country i 's tradable to one in i 's currency, perfect competition then implies

$$w_i = A_{T_i}, \quad P_{N_i} = A_{T_i}/A_{N_i}.$$

Linear production and the producer-price normalization therefore fix wages and domestic producer prices in own currency, so relative international producer prices adjust solely through exchange rates.

We consider nested-CES preferences that allow substitution between domestic and imported tradables to differ from substitution across foreign suppliers:

$$\begin{aligned} C_i &= C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} && \text{(tradable share } \alpha_T), \\ C_{T_i} &= \left[\alpha_D^{1/\eta} C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} && \text{(home weight } \alpha_D, \text{ elasticity } \eta), \\ M_i &= \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} && \text{(origin weights } \beta_{ij}, \text{ elasticity } \rho). \end{aligned}$$

The top layer of the nested utility function is Cobb–Douglas, so the expenditure share on tradables is fixed at α_T and the share on non-tradables at $1 - \alpha_T$. Within tradables, η governs substitution between the domestic tradable and the import composite, while ρ governs substitution across foreign origins. The latter margin is inactive at $N = 2$, when there is only one foreign supplier, and becomes relevant at $N \geq 3$.

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Symmetry across countries does not eliminate home bias: domestic tradables retain weight α_D . Full symmetry between the domestic tradable and each foreign tradable would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which we do not impose.

As in Section 2, let e_{ij} denote units of i 's currency per unit of j 's, and take A 's currency as the numeraire, with $\nu_j \equiv \log e_{Aj}$ and $d\nu_j = d \log e_{Aj}$. We allow for general ad valorem tariffs $\tau_{ij} \geq 0$ levied by country i on imports from country j , with $\tau_{ii} = 0$. Since the producer price of country j 's tradable is normalized to one in j 's currency, its consumer price in country i is $p_{ij} = (1 + \tau_{ij})e_{ij}$, while $p_{ii} = 1$. The corresponding import, tradable, and aggregate price indices are

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1-\alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T}. \quad (17)$$

The associated tradable expenditure shares of income are

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad s_{ij} = \alpha_T (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i. \quad (18)$$

Thus $s_{ij}I_i$ is tariff-inclusive expenditure on imports from j , while $s_{ij}I_i/(1 + \tau_{ij})$ is the corresponding tariff-exclusive border value.

Household income consists of labor income, $w_i L_i$, and tariff revenue, which is rebated lump-sum:

$$I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (19)$$

At the free-trade baseline, $I_i = w_i L_i$. The Cobb–Douglas top layer then fixes expenditure, and hence equilibrium labor, shares at α_T in tradables and $1 - \alpha_T$ in non-tradables, independently of exchange rates and the lower-level substitution elasticities. Exchange-rate movements alone therefore do not reallocate labor between the tradable and non-tradable sectors. Tariffs are different because rebated tariff revenue raises household income relative to labor income. The resulting increase in expenditure on non-tradables requires labor to shift toward the non-tradable sector. This income channel will matter below for changes in average protection; at the symmetric free-trade benchmark, pure changes in relative treatment leave tariff revenue, income, and this sectoral allocation unchanged to first order.

To express bilateral flows in a common currency, define $\tilde{I}_i \equiv e_{Ai} I_i$. Since tariff revenue is a domestic transfer, trade balances are measured at tariff-exclusive border prices:

$$\text{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \quad f_{ij} \equiv \frac{s_{ij} \tilde{I}_i}{1 + \tau_{ij}}, \quad (20)$$

where f_{ij} is the border value, in A 's currency, of country i 's imports from country j . As in Section 2, Walras' law leaves $N - 1$ independent trade-balance conditions. With no internationally traded assets, equilibrium requires

$$\text{TB}_i(\{e_{Aj}\}_{j \neq A}, \{\tau_{jk}\}_{j \neq k}) = 0, \quad i \neq A,$$

which, for a given set of tariffs, determine the $N - 1$ equilibrium exchange rates relative to A .

Define country i 's nominal effective exchange rate using baseline bilateral trade weights as

$$E_i \equiv \prod_{j \neq i} e_{ij}^{w_{ij}}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})}, \quad (21)$$

where the weights are evaluated at the baseline equilibrium. At the symmetric benchmark, $w_{ij} = 1/(N - 1)$, so the log nominal effective exchange rate is the simple average of country i 's bilateral log exchange rates.

Finally, define bilateral terms of trade and real exchange rates as

$$\text{ToT}_{ij} \equiv \frac{1}{e_{ij}}, \quad q_{ij} \equiv \frac{e_{ij}P_j}{P_i}. \quad (22)$$

An appreciation of country i against country j therefore corresponds to an improvement in i 's bilateral terms of trade.

3.2 The two-country case

Appendix B gives the general N -country linearization, including the $N = 2$ calculations used here. Write $m \equiv \alpha_T(1 - \alpha_D)$ for the import share of income, and suppose A raises its tariff on B by $d\tau$. With only one foreign supplier, ρ is inactive. Define the domestic diversion margin

$$\rho_D \equiv \underbrace{\alpha_D \eta}_{\text{domestic-tradable substitution}} + \underbrace{(1 - \alpha_D)(1 - \alpha_T)}_{\text{tariff-rebate/non-tradable channel}} > 0, \quad (23)$$

and the exchange-rate adjustment margin

$$\mathcal{M}_2 \equiv \underbrace{1}_{\text{valuation}} + \underbrace{2\alpha_D(\eta - 1)}_{\text{two-sided trade adjustment}}. \quad (24)$$

Linearizing (20) at the balanced free-trade baseline gives

$$d\text{TB}_A = m\mathcal{M}_2 d\nu_B + m\rho_D d\tau = 0. \quad (25)$$

Thus ρ_D determines the trade-balance disturbance generated by the tariff at unchanged exchange rates, while $\mathcal{M}_2$ determines the trade-balance response to the exchange rate at unchanged tariffs. Equilibrium therefore requires

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho_D}{\mathcal{M}_2}. \quad (26)$$

The two margins have different economic origins. The tariff directly raises the price of imports in A , shifting expenditure toward the domestic economy. The first term in (23), $\alpha_D \eta$, captures substitution from imports toward the domestic tradable. The second, $(1 - \alpha_D)(1 - \alpha_T)$, comes from the income effect of rebated tariff revenue. As discussed in Section 3.1, the rebate raises household income and expenditure on non-tradables, shifting labor toward the non-tradable sector. Thus ρ_D combines substitution toward the domestic tradable with the tariff-rebate channel operating through non-tradables. In particular, $\text{TB}_{A\tau} = m\rho_D > 0$, so the tariff unambiguously improves A 's trade balance at unchanged exchange rates.

The exchange-rate adjustment margin $\mathcal{M}_2$ instead reflects both sides of the bilateral trade relationship. A change in e_{AB} affects A 's demand for imports from B and B 's demand for exports from A . At the symmetric benchmark, the same domestic-import substitution margin appears on both sides, giving the factor 2 in $2\alpha_D(\eta - 1)$. Unlike a tariff change, an exchange-rate movement at unchanged tariffs does not generate tariff revenue and therefore does not activate the income

channel toward non-tradables. Indeed,

$$\varepsilon_X = \varepsilon_M = 1 + \alpha_D(\eta - 1),$$

so

$$\mathcal{M}_2 = \varepsilon_X + \varepsilon_M - 1. \quad (27)$$

The remaining 1 is the usual valuation term in the Marshall–Lerner expression. Hence the Marshall–Lerner condition of Section 2.1 is simply $\mathcal{M}_2 > 0$, which holds whenever $\eta \geq 1$.

Equation (26) therefore has the same disturbance-adjustment interpretation as Section 2.1: the tariff creates a trade-balance disturbance governed by the domestic diversion margin ρ_D , and the exchange rate restores balance with strength governed by the adjustment margin $\mathcal{M}_2$. Appendix A gives the exact finite-tariff Cobb–Douglas solution.

3.3 The N -country case

Now consider the N -country case, where A 's trading partners are symmetric in the sense of Section 2.4. Proposition 2.3 shows that any tariff change can be decomposed into average protection and relative treatment, with corresponding common and relative exchange-rate responses. The nested-CES structure now gives economic content to these two components and shows how the disturbance-adjustment logic of the two-country case extends to a multilateral setting.

Theorem 3.1 (Exchange-rate responses in the symmetric nested-CES model). *Consider the nested-CES model of Section 3 at the symmetric benchmark. Let $\mathbf{t} = (t_j)_{j \neq A}$ denote any vector of tariff changes imposed by A , with average $\bar{t}$ as in (11). Define the multilateral exchange-rate adjustment margin*

$$\mathcal{M}_N \equiv \frac{1}{N-1} \left[\underbrace{1}_{\text{valuation}} + \underbrace{N\alpha_D(\eta-1)}_{\text{domestic-import adjustment}} + \underbrace{(N-2)\rho}_{\text{foreign-origin adjustment}} \right]. \quad (28)$$

The MML condition (8) holds if and only if $\mathcal{M}_N > 0$. Since $\alpha_D > 0$ and $\rho > 0$, $\eta \geq 1$ is sufficient for $\mathcal{M}_N > 0$.

When $\mathcal{M}_N > 0$, the average-protection and relative-treatment coefficients are

$$\lambda_A = -\frac{\rho_D}{\mathcal{M}_N}, \quad \lambda_R = -\frac{\rho}{N\mathcal{M}_N},$$

and the equilibrium exchange-rate response is

$$d\nu_j = \lambda_A \bar{t} + \lambda_R \tilde{t}_j. \quad (29)$$

For $N = 2$, there is only one trading partner, so relative treatment is mechanically zero and

$$\mathcal{M}_2 = 2\alpha_D(\eta - 1) + 1.$$

Theorem 3.1 therefore reduces directly to the two-country result in Section 3.2. The same two

objects continue to govern the common component of the response: ρ_D measures the trade-balance disturbance generated by an increase in overall protection at unchanged exchange rates, while $\mathcal{M}_N$ measures the adjustment of A 's aggregate trade balance to a common movement in its exchange rates.

To see the latter interpretation, consider a common exchange-rate change $d\nu_j = d\nu_A^E$ for all $j \neq A$, holding tariffs fixed. At the symmetric benchmark,

$$dTB_A = m\mathcal{M}_N d\nu_A^E. \quad (30)$$

Thus $\mathcal{M}_N$ is the multilateral counterpart of $\mathcal{M}_2$: up to the import-share scale factor m , it gives the response of A 's trade balance vis-a-vis the rest of the world to its effective exchange rate. The additional elasticity ρ enters when $N > 2$ because the exchange rate now affects a richer export-side substitution margin. In the two-country case, changes in e_{AB} affect both A 's imports from B and A 's exports to B through the domestic-import substitution elasticity η . With additional countries, foreign buyers can also substitute between goods from A and goods from other foreign suppliers, so substitution across foreign origins, governed by ρ , also contributes to the exchange-rate adjustment margin.

Average protection preserves the same tariff-side intuition as in the two-country case. A common increase in tariffs shifts expenditure toward the domestic economy. Direct substitution toward the domestic tradable is governed by η , while rebated tariff revenue raises income and expenditure on non-tradables, shifting labor toward the non-tradable sector. These effects are summarized by the same domestic diversion margin ρ_D defined in (23). Combining this tariff-induced disturbance with the aggregate exchange-rate adjustment in (30) gives

$$\lambda_A = -\frac{\rho_D}{\mathcal{M}_N}.$$

Average protection therefore affects A 's effective exchange rate through exactly the same disturbance-adjustment mechanism as in the two-country case.

Relative treatment operates differently. A pure relative-treatment change satisfies $\mathbf{1}'\tilde{\mathbf{t}} = 0$ and therefore leaves average protection, tariff revenue, and household income unchanged to first order at the symmetric free-trade benchmark. It consequently leaves the allocation of labor between tradables and non-tradables unchanged. Instead, it reallocates expenditure across foreign suppliers, so the tariff-induced disturbance in the relative direction is governed by ρ . The corresponding relative exchange-rate movement also changes competitiveness across foreign suppliers and therefore provides a stronger adjustment margin. Under symmetry, this relative adjustment margin is $N\mathcal{M}_N$, yielding

$$\lambda_R = -\frac{\rho}{N\mathcal{M}_N}.$$

Thus the additional factor $1/N$ reflects the stronger adjustment of relative exchange rates under symmetry. A relative exchange-rate movement restores trade balances N times as strongly as a common movement, so a given disturbance across foreign suppliers requires correspondingly less relative exchange-rate adjustment.

The two coefficients are therefore governed by three economically distinct margins. The domestic diversion margin ρ_D determines how strongly average protection shifts expenditure toward the domestic economy; ρ determines how strongly relative treatment reallocates expenditure across foreign suppliers; and $\mathcal{M}_N$ determines how strongly the multilateral trade balance responds to exchange-rate adjustment. For $N \geq 3$, $\lambda_R < 0$, so A appreciates more against partners facing larger tariff increases. The formal derivation, including the underlying eigenvalues of G and $-J$, is given in Appendix B.

3.4 A unilateral tariff

Suppose now that A raises its tariff on B alone, so that $\mathbf{t} = d\tau \mathbf{e}_B$. Then

$$\bar{t} = \frac{d\tau}{N-1}, \quad \tilde{t}_B = \frac{N-2}{N-1}d\tau, \quad \tilde{t}_C = -\frac{d\tau}{N-1} \quad \text{for each bystander } C.$$

Since relative treatment averages to zero, only average protection affects the effective exchange rate. Using $\lambda_A = -\rho_D/\mathcal{M}_N$,

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_D}{(N-1)\mathcal{M}_N} < 0, \quad (31)$$

so a unilateral tariff always appreciates A 's effective exchange rate at the symmetric benchmark, for every $N \geq 2$. The factor $1/(N-1)$ reflects that a tariff on one partner raises average protection by only $d\tau/(N-1)$.

The bilateral responses additionally depend on relative treatment. Against the targeted country, average protection and relative treatment reinforce each other:

$$\frac{d\nu_B}{d\tau} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)\mathcal{M}_N} < 0, \quad (32)$$

so A always appreciates against B , again for every $N \geq 2$. For $N \geq 3$ there is also an untargeted bystander C , for which the two components work in opposite directions:

$$\frac{d\nu_C}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)\mathcal{M}_N}. \quad (33)$$

Average protection shifts expenditure toward the domestic economy and pushes A to appreciate against C , while C 's favorable relative treatment redirects expenditure toward C and pushes in the opposite direction. Hence A depreciates against C if and only if

$$\rho > N\rho_D. \quad (34)$$

The threshold compares the two diversion margins identified in Section 3.3. A larger η strengthens substitution toward the domestic tradable and therefore raises ρ_D , while the tariff-rebate channel toward non-tradables also contributes to ρ_D . Both make a bystander depreciation less likely. A larger ρ , by contrast, strengthens diversion across foreign suppliers and makes it more likely. The factor N reflects the stronger adjustment of relative exchange rates, which gives the relative-

treatment response the additional factor $1/N$ in $\lambda_R = -\rho/(N\mathcal{M}_N)$, so diversion across foreign suppliers must be sufficiently strong to overcome the average-protection effect governed by ρ_D .

In particular, the reversal cannot occur when the two substitution elasticities coincide and $\alpha_D \geq 1/N$. If $\rho = \eta = \sigma$, then

$$N\rho_D - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T), \quad (35)$$

which is nonnegative for $\alpha_D \geq 1/N$. Hence $\rho \leq N\rho_D$ and the reversal condition cannot hold.¹ More generally, for $\alpha_D \geq 1/N$,

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0,$$

so $\rho > \eta$ is necessary, but not sufficient, for a bystander depreciation. Appendix C gives the corresponding real exchange-rate results.

The bystander's trade balance can nevertheless improve at unchanged exchange rates even when its bilateral exchange rate does not reverse. From (49),

$$F_C = \frac{m(\rho - \rho_D)}{(N - 1)^2}, \quad (36)$$

so

$$F_C > 0 \iff \rho > \rho_D, \quad \text{whereas} \quad d\nu_C > 0 \iff \rho > N\rho_D.$$

The first condition concerns only the initial tariff-induced trade-balance disturbance: diversion toward C exceeds diversion toward the domestic economy once $\rho > \rho_D$. The exchange-rate response must also account for how effectively relative exchange-rate movements restore trade balance. Because the relative-treatment response carries the additional factor $1/N$, an improvement in C 's trade balance at unchanged exchange rates need not be large enough to reverse the bilateral exchange-rate response. Figure 2 illustrates the distinction.

¹Equality in (35) requires $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$, $d\nu_C/d\tau = -(1 - \alpha_T)/(N\sigma)$, which approaches zero from below as $\alpha_T \rightarrow 1$; see Appendix B.

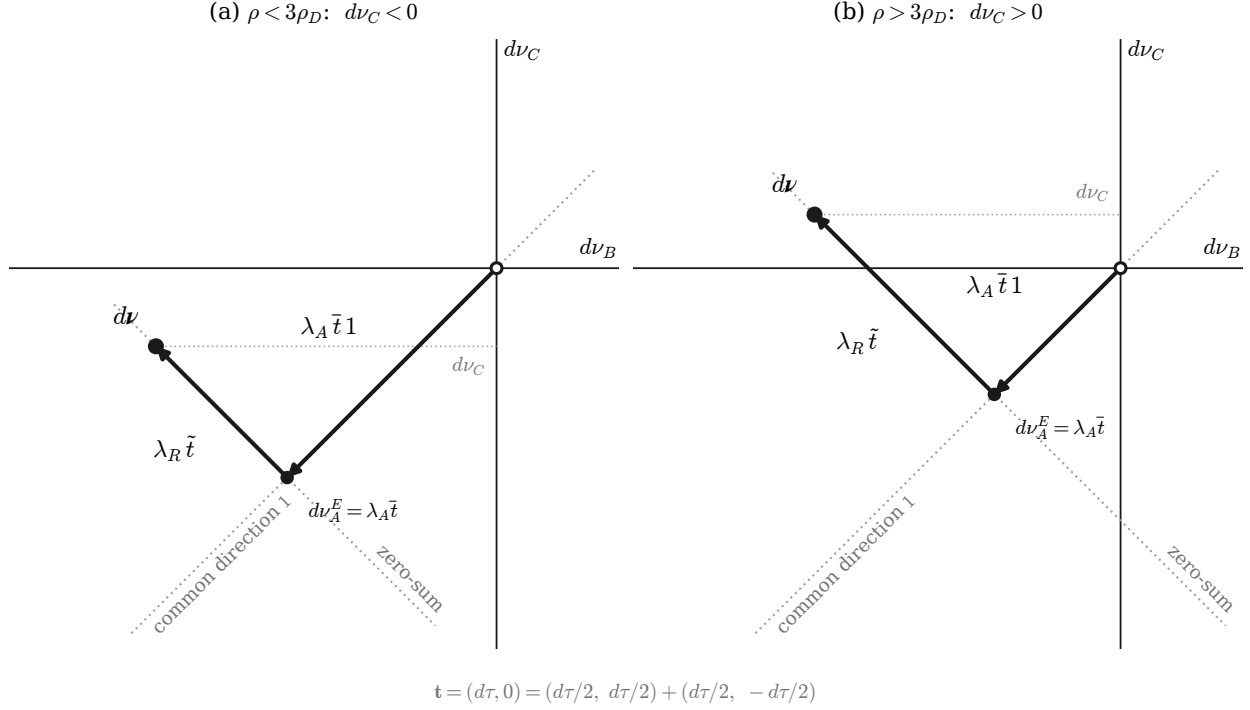


Figure 2: Average protection and relative treatment under a unilateral tariff at $N = 3$. Panel (a): $\rho < 3\rho_D$; panel (b): $\rho > 3\rho_D$.

4 Trade Wars and Preferential Tariffs

Because our analysis is local to a given baseline, simultaneous tariff changes by several countries can be analyzed by considering each country's tariff change separately and then adding the resulting first-order effects. At the symmetric benchmark, each tariffing country's contribution to equilibrium exchange rates follows from Theorem 3.1.

4.1 Trade wars

We first consider bilateral retaliation. Suppose A raises its tariff on B by $d\tau$ and B retaliates by raising its tariff on A by the same amount. By symmetry, the two effects on the A – B bilateral rate cancel. For any bystander C , superposing the two unilateral responses gives

$$\frac{d\nu_B}{d\tau} = 0, \quad \frac{d\nu_C}{d\tau} = \frac{\rho - \rho_D}{(N-1)\mathcal{M}_N}, \quad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \frac{\rho - \rho_D}{(N-1)\mathcal{M}_N}. \quad (37)$$

Relative to a unilateral tariff, B 's tariff on A adds a second relative-treatment effect in C 's favor, which offsets the additional $1/N$ factor in the relative-treatment response. The bystander response therefore changes sign once ρ exceeds ρ_D rather than $N\rho_D$, the same point at which C 's trade balance improves at unchanged exchange rates.

We next consider a multilateral trade war in which A raises its tariff by $d\tau$ on every partner and every partner retaliates by raising its tariff by $d\tau$ on A . This configuration can be viewed

as the sum of $N - 1$ bilateral retaliations involving A . For any A - j bilateral rate, the bilateral retaliation between A and j contributes zero by (37), while each of the remaining $N - 2$ retaliations contributes the same bystander response.

Proposition 4.1 (Retaliation and trade wars). *At the symmetric benchmark of the nested-CES model, with $N \geq 3$ and under MML:*

- (i) *Under bilateral retaliation between A and B , $d\nu_B = 0$. Both countries depreciate against every bystander, and hence in effective terms, if and only if $\rho > \rho_D$. Exchange rates among bystanders are unchanged.*
- (ii) *In a multilateral trade war in which A tariffs every partner and every partner retaliates against A by the same amount,*

$$\frac{d\nu_j}{d\tau} = \frac{d\nu_A^E}{d\tau} = \frac{(N-2)(\rho - \rho_D)}{(N-1)\mathcal{M}_N} \quad \forall j \neq A, \quad (38)$$

while exchange rates among A 's partners are unchanged. Hence A depreciates against every partner if and only if $\rho > \rho_D$.

The two configurations differ in the pattern of bilateral adjustment. Under bilateral retaliation the A - B rate is unchanged and adjustment occurs against the untariffed bystanders, whereas in the multilateral trade war there are no untariffed bystanders and all of A 's bilateral rates move together.

4.2 Preferential tariffs and competing exporters

The bystander response in Section 3.4 is closely related to the third-country terms-of-trade effect studied in the preferential-trade-agreement literature. We use the competing-exporter model as a benchmark for that effect, first deriving its signs and then comparing them with our results.

Following Viner (1950) and Mundell (1964), consider one importing country and two competing exporters. Importer A is endowed only with the numeraire good y , while exporters B and C are endowed with good x . Preferences are quasilinear, $U_i = u_i(x_i) + y_i$. Country A levies specific tariffs t_B and t_C . Let p be the domestic price of x in A , and let q_j be exporter j 's price of x in terms of the numeraire, which is j 's terms of trade. Market clearing requires

$$p = q_B + t_B = q_C + t_C, \quad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \quad M'_A < 0, \quad E'_j > 0, \quad (39)$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$. Differentiating with respect to t_B gives

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1), \quad \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (40)$$

Thus raising the tariff on B worsens B 's terms of trade and improves those of the competing exporter C . Reversing the tariff change also gives the familiar preferential-liberalization result,

where lowering the tariff on B improves B 's terms of trade and worsens those of the excluded country C .

In our model, $\text{ToT}_{CA} = e_{AC}$ by (22), so an improvement in C 's terms of trade corresponds to an increase in e_{AC} . Whereas the competing-exporter model delivers this improvement for any admissible demand and supply elasticities, at the symmetric free-trade benchmark of the nested-CES model (34) shows that e_{AC} increases only if $\rho > N\rho_D$. By the same condition, preferential liberalization toward B , $\mathbf{t} = -d\tau \mathbf{e}_B$, worsens the excluded country's terms of trade only if $\rho > N\rho_D$.²

The difference arises because, in the competing-exporter benchmark, A does not produce the importable, so a tariff on B can only shift demand toward C , whereas in our model part of that demand is diverted toward A 's own tradable, with the associated income effects, so the relevant comparison is with the domestic diversion margin ρ_D . The third-country effect therefore survives only when diversion across foreign suppliers, governed by ρ , is strong enough relative to ρ_D , with the factor N reflecting the stronger adjustment of relative exchange rates rather than the initial trade-balance disturbance.

The two environments are not formally nested. The competing-exporter model features a common importable supplied by B and C under quasilinear preferences, whereas our model has country-specific differentiated tradables and nested-CES preferences. Related competing-exporter models include Bagwell and Staiger (1999) and Ornelas (2005), while Krugman (1991) and Bond and Syropoulos (1996) study related bloc and endowment environments. Once the importing country can also substitute toward its own goods, the standard third-country terms-of-trade effect holds only when substitution across foreign suppliers is sufficiently strong.

5 Conclusion

This paper begins from the observation that, to first order, a tariff changes countries' trade balances at unchanged exchange rates, summarized by F , after which exchange rates adjust through $(-J)^{-1}$. In the two-country case, this gives the conventional wisdom that a tariff improves the trade balance at unchanged exchange rates and, under the Marshall–Lerner condition, leads to an appreciation of the tariffing country's currency. Going beyond this conventional wisdom, we use this framework to examine how the exchange-rate response changes in a multilateral setting, where expenditure can also be redirected toward other foreign suppliers and the resulting exchange-rate adjustment depends on the full pattern of trade-balance disturbances across countries.

Under symmetry, we show that the impact of a general tariff vector can be decomposed into an average-protection component and a relative-treatment component. Average protection determines the effective exchange-rate response, while relative treatment determines deviations of bilateral rates from that common response. In the nested-CES model, the tariff disturbances along these two components are governed by ρ_D and ρ , while exchange-rate adjustment is governed by the multilateral adjustment margin $\mathcal{M}_N$. At the symmetric benchmark of the nested-CES model, the

²While this captures the mechanism behind preferential liberalization, the closed-form threshold is derived around free trade rather than around a positive common external tariff.

tariffing country still appreciates against the targeted partner, but its exchange rate against an untariffed partner can move in either direction. In particular, a unilateral tariff always appreciates the tariffing country's effective exchange rate at this benchmark, while the bilateral rate against an untariffed bystander depreciates when substitution across foreign suppliers is sufficiently strong relative to the domestic diversion margin ρ_D , which combines substitution toward the domestic tradable with the income effect of rebated tariff revenue.

We also examine how these results change under retaliation and compare them with the competing-exporter benchmark. Several extensions are natural. Allowing for asymmetric trade shares, richer production structures, and asset trade would help assess how the mechanisms identified here operate away from the symmetric balanced-trade benchmark, while quantitative applications could evaluate their importance for observed tariff changes and exchange-rate movements. More generally, the analysis shows that once tariffs differ across trading partners, bilateral exchange-rate responses need not follow the familiar two-country result even when the effective exchange rate does.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$, expenditure shares are fixed at their weights. From (19), $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$, and $TB_A = 0$ becomes $e m w_B L_B = m I_A / (1 + \tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}, \quad \left. \frac{d \log e}{d\tau} \right|_{\tau=0} = -(1 - m) = -\left. \frac{\rho_D}{\mathcal{M}_2} \right|_{\eta=1}. \quad (41)$$

Thus $e(\tau)$ is decreasing in τ , so the sign of the linearized response in (26) is exact for all tariff levels in this case. Here $\eta = 1$ fixes the elasticity, not the weight α_D .

B The Linearized System

At the balanced free-trade baseline, let $m \equiv \alpha_T(1 - \alpha_D)$, $b_{ij} \equiv \beta_{ij}$, and $f_{ij} = s_{ij}\tilde{I}_i$, with $\sum_k f_{ki} = \sum_j f_{ij}$. All incomes and flows are measured in the common currency. Linearizing the demand, income, and trade-balance equations gives

$$dp_{ij} = d\nu_j - d\nu_i + d\tau_{ij} \quad (dp_{ii} = 0), \quad (42)$$

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}, \quad (43)$$

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (44)$$

$$d \log s_{ij} = (1 - \rho)(dp_{ij} - d \log P_{M_i}) + (1 - \eta)\alpha_D d \log P_{M_i}, \quad (45)$$

$$d \log \tilde{I}_i = d\nu_i + \sum_{j \neq i} s_{ij} d\tau_{ij}, \quad (46)$$

$$d \text{TB}_i = \sum_{k \neq i} f_{ki} (d \log s_{ki} + d \log \tilde{I}_k - d\tau_{ki}) - \sum_{j \neq i} f_{ij} (d \log s_{ij} + d \log \tilde{I}_i - d\tau_{ij}). \quad (47)$$

These equations determine the model-specific matrices J and G in (4). At $N = 2$ they reduce to the calculation in Section 3.2.

$$\tilde{J}_{il} = \frac{m\mathcal{M}_N}{N-1} \quad (l \neq i), \quad (48)$$

with $\mathcal{M}_N$ as in (28). Lemma B.1 below gives the derivation.

Lemma B.1 (Full Jacobian at the symmetric baseline). *At the symmetric balanced baseline of Appendix B, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full $N \times N$ Jacobian of the nested-CES model is*

$$\tilde{J}_{il} = \frac{m\mathcal{M}_N}{N-1} \quad (l \neq i), \quad \tilde{J}_{ii} = -m\mathcal{M}_N, \quad \mathcal{M}_N = \frac{N\alpha_D(\eta-1) + (N-2)\rho + 1}{N-1}.$$

Proof. Set $d\tau_{jk} = 0$, so $dp_{ij} = d\nu_j - d\nu_i$ and $d \log \tilde{I}_i = d\nu_i$. Writing $S \equiv \sum_j d\nu_j$, (43) gives

$$d \log P_{M_k} = \frac{S - N d\nu_k}{N-1}, \quad \frac{\partial \log P_{M_k}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side,

$$\sum_{j \neq i} (dp_{ij} - d \log P_{M_i}) = 0,$$

so the $(1 - \rho)$ terms cancel when (45) is summed over $j \neq i$. Thus,

$$\sum_{j \neq i} d \log s_{ij} = (N-1)(1 - \eta)\alpha_D d \log P_{M_i}.$$

Substituting into (47) gives

$$dTB_i = \frac{m}{N-1} \left[\sum_{k \neq i} (d \log s_{ki} + d\nu_k) - (N-1)(1-\eta)\alpha_D d \log P_{M_i} - (N-1) d\nu_i \right].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l} \sum_{k \neq i} \log s_{ki} = -(1-\rho) \frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms gives

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} \left[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1) \right] = \frac{m\mathcal{M}_N}{N-1}.$$

The diagonal follows from (10):

$$\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -m\mathcal{M}_N.$$

At $N = 2$ this gives $\tilde{J}_{AB} = m\mathcal{M}_2$, the coefficient in (25); deleting row and column A at $N = 3$ reproduces J in (56). $\square$

The two-country case. At the balanced free-trade baseline with normalized incomes, $s_{AB} = s_{BA} = m$. At $N = 2$ each country has a single import origin, so $dp_{AB} = d \log P_{M_A} = d\nu_B + d\tau$ and $dp_{BA} = d \log P_{M_B} = -d\nu_B$. Log-differentiating (18) and (19), with incomes in the common currency,

$$\begin{aligned} d \log s_{AB} &= (1-\eta)\alpha_D(d\nu_B + d\tau), & d \log s_{BA} &= -(1-\eta)\alpha_D d\nu_B, \\ d \log \tilde{I}_A &= m d\tau, & d \log \tilde{I}_B &= d\nu_B. \end{aligned}$$

Substituting into (20) and using that the tariff wedge lowers the border value of A 's imports by $d\tau$,

$$dTB_A = m \left[(1 + 2\alpha_D(\eta-1))d\nu_B + (1 - m + \alpha_D(\eta-1))d\tau \right],$$

which is (25), with $\mathcal{M}_2 = 1 + 2\alpha_D(\eta-1)$ and $\rho_D = 1 - m + \alpha_D(\eta-1)$, and (26) follows. For the elasticity form, hold income fixed and let $M(p)$ denote import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 - (1-\eta)\alpha_D = 1 + \alpha_D(\eta-1).$$

By symmetry, $\varepsilon_X = 1 + \alpha_D(\eta-1)$. Differentiating $TB_A(e, \tau) = X(1/e) - eM((1+\tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives

$$\frac{\partial TB_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1).$$

Hence $\varepsilon_X + \varepsilon_M - 1 = 2\alpha_D(\eta - 1) + 1 = \mathcal{M}_2$, so the stability condition $\mathcal{M}_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$. Since $\rho_D > 0$ by (23), the slope $-\rho_D/\mathcal{M}_2$ is negative whenever $\mathcal{M}_2 > 0$.

The tariff disturbance and the four eigenvalues. At unchanged exchange rates, the effect on partner i 's trade balance follows from (47). Since only i 's exports to A are directly affected,

$$d\text{TB}_i|_{\nu \text{ fixed}} = f_{Ai}(d \log s_{Ai} + d \log \tilde{I}_A - t_i) = \frac{m}{N-1} \left[-\rho_D \bar{t} - \rho(t_i - \bar{t}) \right]. \quad (49)$$

Reading off the coefficients on $\bar{t}$ and $\tilde{t}_i$ gives the tariff-effect eigenvalues

$$g_S = -\frac{m\rho_D}{N-1}, \quad g_D = -\frac{m\rho}{N-1}. \quad (50)$$

Deleting row and column A from (48) gives the exchange-rate adjustment matrix

$$-J = c(NI - \mathbf{1}\mathbf{1}'), \quad c \equiv \frac{m\mathcal{M}_N}{N-1}, \quad (51)$$

whose eigenvalues are $j_S = c$ on $\text{span}\{\mathbf{1}\}$ and $j_D = Nc$ on the zero-sum subspace:

$$j_S = \frac{m\mathcal{M}_N}{N-1}, \quad j_D = \frac{Nm\mathcal{M}_N}{N-1}. \quad (52)$$

Dividing (50) by (52) gives the coefficients of Theorem 3.1,

$$\lambda_A = \frac{g_S}{j_S} = -\frac{\rho_D}{\mathcal{M}_N}, \quad \lambda_R = \frac{g_D}{j_D} = -\frac{\rho}{N\mathcal{M}_N}.$$

The unilateral tariff. For $\mathbf{t} = d\tau \mathbf{e}_B$, $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$. By Theorem 3.1,

$$d\nu_B = \frac{\lambda_A + (N-2)\lambda_R}{N-1} d\tau = -\frac{N\rho_D + (N-2)\rho}{N(N-1)\mathcal{M}_N} d\tau, \quad d\nu_C = \frac{\lambda_A - \lambda_R}{N-1} d\tau = \frac{\rho - N\rho_D}{N(N-1)\mathcal{M}_N} d\tau,$$

while $d \log e_{BC} = d\nu_C - d\nu_B = -\lambda_R d\tau$ and $d\nu_A^E = \lambda_A d\tau/(N-1) = -\rho_D d\tau/[(N-1)\mathcal{M}_N]$. Since $\rho_D > 0$ by (23) and $\mathcal{M}_N > 0$ under MML, these give (32)–(34).

Derivation of (35). With $\rho = \eta = \sigma$,

$$N\rho_D = N[1 + \alpha_D(\sigma - 1) - \alpha_T(1 - \alpha_D)],$$

so

$$N\rho_D - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

For $\alpha_D \geq 1/N$ both terms are nonnegative, with both zero only at $\alpha_D = 1/N$, $\alpha_T = 1$; hence $\rho = \sigma \leq N\rho_D$. Since ρ_D does not involve ρ ,

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

At $\alpha_D = 1/N$,

$$N\rho_D - \sigma = (N-1)(1 - \alpha_T), \quad \mathcal{M}_N = \sigma,$$

so by (33),

$$\frac{d \log e_{AC}}{d\tau} = -\frac{1 - \alpha_T}{N\sigma}.$$

Relative treatment across partners. By (29), $d\nu_j - d\nu_k = \lambda_R(t_j - t_k)$ with $\lambda_R < 0$, so $t_j > t_k$ implies $d\nu_j < d\nu_k$. With $\bar{t} > 0$, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_D \bar{t} \quad \Longleftrightarrow \quad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_D}{\rho}. \quad (53)$$

To see this, note that $d\nu_j > 0$ if and only if $\lambda_A \bar{t} + \lambda_R(t_j - \bar{t}) > 0$. Substituting λ_A and λ_R from Theorem 3.1 and multiplying by $\mathcal{M}_N > 0$ gives $-\rho_D \bar{t} - \frac{\rho}{N}(t_j - \bar{t}) > 0$, which rearranges to (53). The right-hand side of the second form is positive only if $\rho > N\rho_D$, so depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_D$ and a sufficiently large tariff differential.

Tariffs on several partners. If A raises its tariff by $d\tau$ on a set of k partners, $1 \leq k \leq N-1$, then for every untargeted partner C and every targeted partner B ,

$$\frac{d \log e_{AC}}{d\tau} = \frac{k(\rho - N\rho_D)}{N(N-1)\mathcal{M}_N}, \quad \frac{d \log e_{AB}}{d\tau} = -\frac{Nk\rho_D + (N-1-k)\rho}{N(N-1)\mathcal{M}_N}, \quad \frac{d\nu_A^E}{d\tau} = -\frac{k\rho_D}{(N-1)\mathcal{M}_N}. \quad (54)$$

For every $k \leq N-2$, the bystander response has the same threshold $\rho > N\rho_D$. When all partners face the same tariff,

$$\frac{d \log e_{Aj}}{d\tau} = -\frac{\rho_D}{\mathcal{M}_N} = \frac{d\nu_A^E}{d\tau} \quad \forall j. \quad (55)$$

All bilateral rates then move together, and substitution across foreign origins plays no role.

To derive these, let $t_j = d\tau$ on the k targets and zero otherwise, so that

$$\bar{t} = \frac{k d\tau}{N-1}, \quad \tilde{t}_B = \frac{N-1-k}{N-1} d\tau$$

for a target, and

$$\tilde{t}_C = -\frac{k d\tau}{N-1}$$

for a bystander. Substituting into (29) gives

$$d\nu_C = \frac{k d\tau}{N-1}(\lambda_A - \lambda_R) = \frac{k d\tau}{N(N-1)\mathcal{M}_N}(\rho - N\rho_D),$$

together with the target and NEER responses in (54). For every $k \leq N-2$, the sign of the bystander response is therefore determined by $\rho - N\rho_D$. At $k = N-1$ there is no bystander and $\tilde{t} = 0$, giving (55).

B.1 Three-country illustration

At $N = 3$, (51) and (49) give the unilateral-tariff system explicitly:

$$J = -\frac{m\mathcal{M}_3}{2} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_D \\ \rho_D - \rho \end{pmatrix}, \quad \mathcal{M}_3 = \frac{3\alpha_D(\eta - 1) + \rho + 1}{2}. \quad (56)$$

With $\mathcal{M}_3 > 0$, (32)–(33) read

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_D}{6\mathcal{M}_3} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_D}{6\mathcal{M}_3}}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{\rho}{3\mathcal{M}_3} > 0. \quad (57)$$

Thus A appreciates against B , B depreciates against C , and A depreciates against C if and only if $\rho > 3\rho_D$. By (9), $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_D$, so the sign of C 's own disturbance F_C , which is positive once $\rho > \rho_D$, does not settle the sign of $d\nu_C$.

Figure 3 plots the two equilibrium conditions,

$$\text{TB}_B = 0 : \quad d\nu_C = 2 d\nu_B + \frac{\rho + \rho_D}{2\mathcal{M}_3} d\tau, \quad \text{TB}_C = 0 : \quad d\nu_C = \frac{1}{2} d\nu_B + \frac{\rho - \rho_D}{4\mathcal{M}_3} d\tau. \quad (58)$$

In the intermediate case $\rho_D < \rho < 3\rho_D$, C gains from trade diversion at unchanged exchange rates and the $\text{TB}_C = 0$ locus shifts up. But the larger shift of $\text{TB}_B = 0$ drives $d\nu_B$ down, and the resulting movement along the upward-sloping $\text{TB}_C = 0$ locus more than offsets that shift: A still appreciates against C .

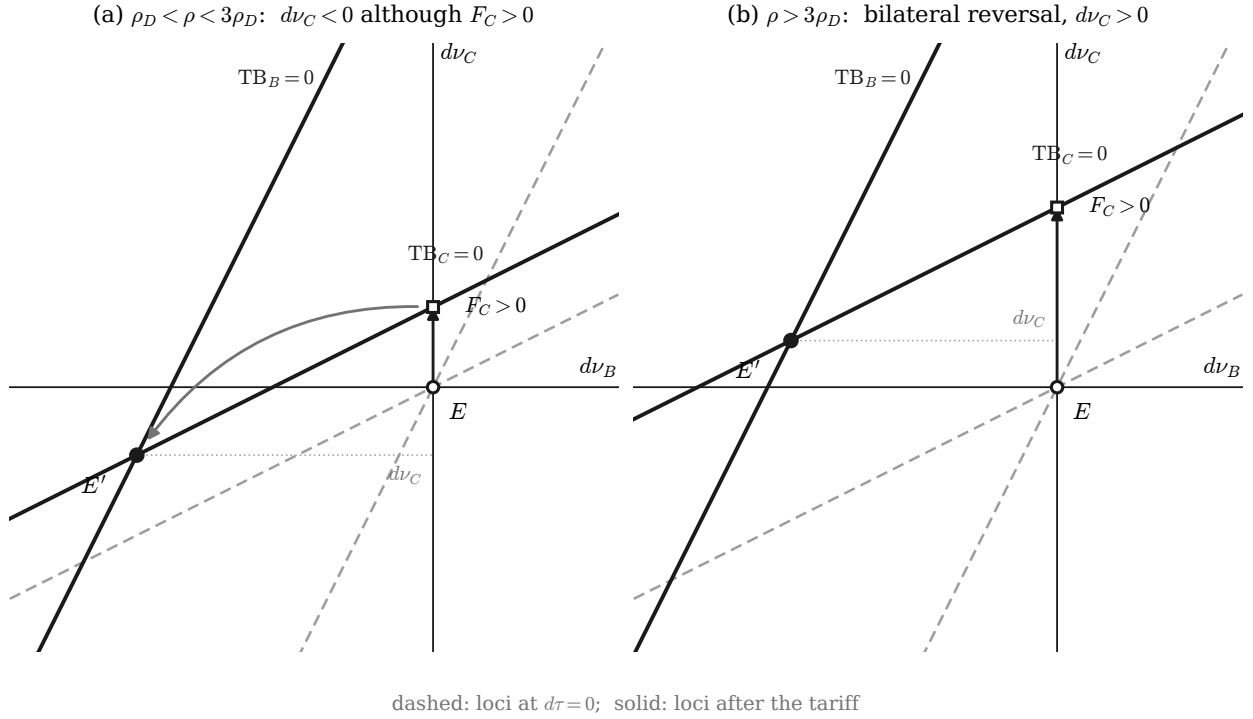


Figure 3: Three-country equilibrium adjustment. The loci are (58); dashed lines are the pre-tariff loci through the initial equilibrium E , solid lines the post-tariff loci intersecting at E' . The open square is the value of dv_C that would clear C 's trade balance at $dv_B = 0$; the curved arrow in panel (a) is the subsequent movement along $TB_C = 0$. Panel (a): $\rho_D < \rho < 3\rho_D$. Panel (b): $\rho > 3\rho_D$. Drawn at $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$, with $\rho = 2.2$ in panel (a) and $\rho = 5$ in panel (b), so that $\rho_D = 1.17$ and $3\rho_D = 3.51$.

C Real exchange rates

This appendix gives the real exchange-rate counterparts of the results in Sections 3.3–3.4. Real effective exchange-rate changes use the weights in (21), $dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij}$.

From (22) and (17),

$$d \log q_{Aj} = d\nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}).$$

For a tariff vector $\mathbf{t}$, symmetry implies

$$d \log P_{M_A} = d\nu_A^E + \bar{t}, \quad d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} d\nu_j$$

for a partner that imposes no tariff. Hence

$$\begin{aligned} d \log q_{Aj} &= \left(1 - \frac{mN}{N-1}\right) d\nu_j - m \bar{t}, \\ dq_A^E &= \left[\left(1 - \frac{mN}{N-1}\right) \lambda_A - m\right] \bar{t}, \\ d \log q_{Aj} - dq_A^E &= \left(1 - \frac{mN}{N-1}\right) \lambda_R \tilde{t}_j. \end{aligned} \tag{59}$$

Thus average protection determines the real effective exchange-rate response, while relative treatment determines bilateral deviations around it, subject to the price-index adjustment in (59).

For a unilateral tariff, $\bar{t} = d\tau/(N-1)$ and

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) d\nu_j - \frac{m}{N-1} d\tau, \quad j = B, C. \tag{60}$$

Setting $d \log q_{AC} = 0$ and using (33), for $m < 1/N$ the real bilateral reversal threshold is

$$\rho_q^*(N) = \frac{[(N-1) - mN] N \rho_D + mN [N \alpha_D (\eta - 1) + 1]}{(N-1)(1 - mN)}. \tag{61}$$

Averaging (60) gives

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_D}{(N-1)\mathcal{M}_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \left(m < \frac{N-1}{N}\right). \tag{62}$$

Proposition C.1 (Real exchange rates). *At the symmetric benchmark of the nested-CES model, consider a unilateral tariff.*

- (i) *If $m < 1/N$ and MML holds at $\rho = \rho_q^*(N)$, the real bilateral reversal threshold exceeds the nominal threshold, $\rho_q^*(N) > N \rho_D$.*
- (ii) *If MML holds and $m < (N-1)/N$, the REER appreciates, $dq_A^E/d\tau < 0$.*

Both results reflect the direct price-index effect of the tariff. At a given nominal exchange rate, the tariff raises A 's consumer prices, making a real bilateral depreciation harder to obtain and

reinforcing the effective appreciation.

D Asymmetry

The disturbance–adjustment representation (6) and MML (8) do not require symmetry, and at any balanced baseline the linearized system (42)–(47) continues to hold with baseline shares and incomes as coefficients: b_{ij} , the share of origin j in i ’s import bundle; h_i , the home share of i ’s tradable expenditure; and $y_i \equiv \tilde{I}_i$, i ’s income in the common currency. What requires symmetry is the collapse of the response to the two coefficients λ_A and λ_R of Proposition 2.3. Away from symmetry, the common and zero-sum directions are generally no longer eigen-directions of G and $-J$, so changes in relative treatment can affect the effective exchange rate. The unconditional NEER appreciation and the linear thresholds of Section 3 therefore need not survive. This appendix records what does.

Bilateral thresholds. The elasticity ρ enters the share responses (45) only through $(1 - \rho)$, so every entry of J and F is linear in ρ . Each entry of $\text{adj}(-J)$ is therefore a polynomial of degree at most $N - 2$ in ρ , and each numerator $[\text{adj}(-J)F]_i$ in (6) a polynomial of degree at most $N - 1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. Symmetry reduces the problem to the two coefficients of Proposition 2.3, yielding the linear thresholds (34) and (57). At a general asymmetric baseline the $N = 3$ numerator is already quadratic in ρ :

$$\text{sign} \frac{d \log e_{AC}}{d\tau} = \text{sign} (c_2 \rho^2 + c_1 \rho + c_0), \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C. \quad (63)$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. Provided MML continues to hold, $c_2 > 0$ whenever A and C each import from both foreign origins, so sufficiently strong substitution across import origins still reverses e_{AC} . When positive roots exist and MML continues to hold, the response is positive beyond the largest positive root, which equals $3\rho_D$ at the symmetric point. Away from symmetry there may be no positive root or several, so there is no universal bilateral threshold, and for $N \geq 4$ a response may change sign more than once as ρ varies.

E Correspondence across N

All entries are at the symmetric benchmark of the nested-CES model. MML holds if and only if $\mathcal{M}_N > 0$.

object	$N = 2$	$N = 3$	general N
domestic diversion margin ρ_D (23)	$\alpha_D \eta + (1 - \alpha_D)(1 - \alpha_T)$, independent of N		
exchange-rate adjustment margin $\mathcal{M}_N$ (28)	$2\alpha_D(\eta - 1) + 1$	$\frac{3\alpha_D(\eta - 1) + \rho + 1}{2}$	$\frac{N\alpha_D(\eta - 1) + (N - 2)\rho + 1}{N - 1}$
average-protection coefficient λ_A	$-\rho_D/\mathcal{M}_2$	$-\rho_D/\mathcal{M}_3$	$-\rho_D/\mathcal{M}_N$
relative-treatment coefficient λ_R	—	$-\rho/(3\mathcal{M}_3)$	$-\rho/(N\mathcal{M}_N)$
unilateral bystander threshold	none	$\rho > 3\rho_D$	$\rho > N\rho_D$
unilateral NEER response	$-\rho_D/\mathcal{M}_2$	$-\rho_D/(2\mathcal{M}_3)$	$-\rho_D/[(N - 1)\mathcal{M}_N]$
trade-war threshold	—	$\rho > \rho_D$	$\rho > \rho_D$

F Proofs

Throughout, $m \equiv \alpha_T(1 - \alpha_D)$ and all derivatives are evaluated at the balanced free-trade baseline of Appendix B.

Proof of Lemma 2.2. Scaling every currency value e_{Al} by a common factor leaves all relative prices in (42) unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i , by the same factor. Thus TB_i is homogeneous of degree one in the currency values. By Euler's theorem, $\sum_l \tilde{J}_{il} = \text{TB}_i$, which is zero at the balanced baseline, so $\sum_l \tilde{J}_{il} = 0$ for all i . Equivalently, in (47), a common shift in all ν_l affects only the income terms, whose export and import contributions cancel by baseline balance. Summing (20) over i gives $\sum_i \text{TB}_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$ for all l . Under Assumption 1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l \neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives

$$|J_{ii}| - \sum_{l \neq i, A} J_{il} = \tilde{J}_{iA} \geq 0, \quad (64)$$

with strict inequality in every row for which $\tilde{J}_{iA} > 0$.

Thus $-J$ is a Z-matrix with nonpositive off-diagonal entries, nonnegative diagonal entries, weak diagonal dominance, irreducibility, and strict diagonal dominance in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$, where $D = \text{diag}(-J) > 0$ and $B \geq 0$. Diagonal dominance implies that every

row sum of $D^{-1}B$ is at most one and that at least one is strictly below one. With irreducibility, its spectral radius therefore satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \geq 0.$$

Nonsingularity follows from $r < 1$. Moreover,

$$\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0.$$

□

Proof of Proposition 2.3. Let M be an $(N-1) \times (N-1)$ matrix with $PMP' = M$ for every permutation matrix P . Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i, j) to (k, l) , $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence

$$M = aI + b\mathbf{1}\mathbf{1}'.$$

Since $\mathbf{1}'\mathbf{1} = N-1$,

$$M\mathbf{1} = [a + (N-1)b]\mathbf{1},$$

while $Mv = av$ for every v satisfying $\mathbf{1}'v = 0$. This gives (16) for $-J$ and G .

For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D}I + \frac{1}{N-1}\left(\frac{1}{j_S} - \frac{1}{j_D}\right)\mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the constant and zero-sum eigenspaces, with response coefficients $\lambda_A = g_S/j_S$ and $\lambda_R = g_D/j_D$. Writing

$$\mathbf{t} = \bar{t}\mathbf{1} + \tilde{t}, \quad \mathbf{1}'\tilde{t} = 0,$$

gives

$$d\boldsymbol{\nu} = \frac{g_S}{j_S}\bar{t}\mathbf{1} + \frac{g_D}{j_D}\tilde{t},$$

and averaging with equal weights yields

$$d\nu_A^E = \frac{g_S}{j_S}\bar{t}.$$

For (15), the off-diagonal entry of $-J$ is

$$b_J = \frac{j_S - j_D}{N-1},$$

so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, the entries of $(-J)^{-1}$ above are nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or its inverse cannot be elementwise nonnegative. At $N = 2$ the zero-sum subspace is trivial and the condition reduces to $j_S > 0$.

Finally, for any two partners $j, k \neq A$,

$$d \log e_{jk} = d\nu_k - d\nu_j = \lambda_R(\tilde{t}_k - \tilde{t}_j) = \lambda_R(t_k - t_j),$$

which completes (14). $\square$

Proof of Theorem 3.1. At the symmetric benchmark, relabeling A 's partners leaves the reduced system unchanged, so Proposition 2.3 applies. Appendix B derives (48), (50) and (52). By (52), $j_S = m\mathcal{M}_N/(N-1)$ and $j_D = Nj_S$, so $j_S > 0$ if and only if $\mathcal{M}_N > 0$, in which case $j_D > j_S$; by (15), MML is therefore equivalent to $\mathcal{M}_N > 0$. If $\eta \geq 1$ then every term in the numerator of (28) is nonnegative and the last is one, so $\mathcal{M}_N > 0$. Given $\mathcal{M}_N > 0$, dividing (50) by (52) gives the coefficients of Theorem 3.1,

$$\lambda_A = \frac{g_S}{j_S} = -\frac{m\rho_D}{N-1} \cdot \frac{N-1}{m\mathcal{M}_N} = -\frac{\rho_D}{\mathcal{M}_N}, \quad \lambda_R = \frac{g_D}{j_D} = -\frac{m\rho}{N-1} \cdot \frac{N-1}{Nm\mathcal{M}_N} = -\frac{\rho}{N\mathcal{M}_N},$$

both negative since $\rho_D > 0$ by (23) and $\rho > 0$. Equation (29) and the three statements following it are then (13) and (14). $\square$

Proof of Proposition 4.1(i). Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (32) gives

$$d \log e'_{BA} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)\mathcal{M}_N}, \quad d \log e'_{BC} = \frac{\rho - N\rho_D}{N(N-1)\mathcal{M}_N}.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d \log e'_{AB} = \frac{N\rho_D + (N-2)\rho}{N(N-1)\mathcal{M}_N}, \quad d \log e'_{AC} = \frac{\rho}{N\mathcal{M}_N}.$$

By linearity,

$$d \log e_{AB}|^w = 0,$$

and

$$d \log e_{AC}|^w = \frac{\rho - N\rho_D}{N(N-1)\mathcal{M}_N} + \frac{\rho}{N\mathcal{M}_N} = \frac{\rho - \rho_D}{(N-1)\mathcal{M}_N}.$$

In terms of Theorem 3.1, the three components are A 's average protection, A 's relative treatment, and B 's relative treatment:

$$d \log e_{AC}|^w = -\frac{\rho_D}{(N-1)\mathcal{M}_N} + \frac{\rho}{N(N-1)\mathcal{M}_N} + \frac{\rho}{N\mathcal{M}_N} = \frac{\rho - \rho_D}{(N-1)\mathcal{M}_N}.$$

By symmetry between A and B , the same expression applies to $d \log e_{BC}|^w$. Averaging gives

$$\left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_D}{(N-1)\mathcal{M}_N}.$$

Under MML, $\mathcal{M}_N > 0$, so the signs are determined by $\rho - \rho_D$. $\square$

Proof of Proposition 4.1(ii). Superpose the N tariff vectors. A 's tariff vector is uniform,

$$\tilde{t}^{(A)} = d\tau, \quad \tilde{t}^{(A)} = 0,$$

and contributes $\lambda_A d\tau$ to every $d \log e_{Aj}$.

Partner j taxes only A . Its contribution to $d \log e_{Aj}$ is

$$-[\lambda_A + (N-2)\lambda_R] \frac{d\tau}{N-1},$$

while each of the $N-2$ other partners contributes $-\lambda_R d\tau$. Summing,

$$\begin{aligned} d \log e_{Aj} &= \lambda_A d\tau - [\lambda_A + (N-2)\lambda_R] \frac{d\tau}{N-1} - (N-2)\lambda_R d\tau \\ &= \frac{(N-2)(\rho - \rho_D)}{(N-1)\mathcal{M}_N} d\tau. \end{aligned}$$

The expression is the same for every j , so it also equals the NEER response, and all cross rates among A 's partners are unchanged. $\square$

Proof of Proposition C.1. From (22) and (17), only tradable prices move, so

$$d \log q_{Aj} = d\nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}),$$

with

$$d \log P_{T_i}^{\text{own}} = (1 - \alpha_D) d \log P_{M_i}.$$

For a tariff vector $\mathbf{t}$,

$$d \log P_{M_A} = \frac{1}{N-1} \sum_j (d\nu_j + t_j) = d\nu_A^E + \bar{t},$$

while for a partner that imposes no tariff,

$$d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} d\nu_j.$$

Substitution gives (59), and $\bar{t} = d\tau/(N-1)$ gives (60).

For (i), setting $d \log q_{AC} = 0$ and substituting $d\nu_C$ from (33) gives

$$[(N-1) - mN] \frac{\rho - N\rho_D}{N} = m(N-1)\mathcal{M}_N(\rho).$$

For $m < 1/N$, solving for ρ gives (61). The coefficient $(N-1) - mN$ on the left is then positive, and MML at $\rho = \rho_q^*(N)$ gives $\mathcal{M}_N > 0$ on the right, so $\rho_q^*(N) > N\rho_D$. For (ii), averaging (60) and using (31) gives (62); under MML $\mathcal{M}_N > 0$, and for $m < (N-1)/N$ both terms are negative. $\square$